

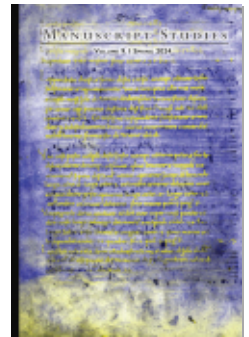


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From Digitization and Images to Text and Content: Transkribus as a Case Study

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From Digitization and Images to Text and Content: Transkribus as a Case Study

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OVER THE PAST DECADES, libraries and archives have been increasingly investing in the digitization of their collections, including manuscripts, rare books, newspapers, and archival material. Many of these resources are freely accessible. However, the accessible material consists only of the metadata of the resources along with their images. The textual content of the resulting digital images is not yet visible, and those seeking to find the content of the resources must study and transcribe individual passages.

The National Library of Israel (NLI), like many national libraries, is involved in some comprehensive digitization projects.¹ One (first launched

I would like to express my sincere gratitude to Sinai Rusinek for her invaluable guidance and assistance throughout the entire process of this study. I am truly grateful for her dedication and contributions to the project.

1 Starting with the pioneer project, the Ketubot Collection (<https://www.nli.org.il/en/discover/manuscripts/ketubot>) began in the late twentieth century with the support of the David and Fela Shapell Digitization Project. It continued with other projects such as Jerusalem Maps, Recorded Treasures of the National Sound Archives, Newton Manuscripts, Historical Jewish Press, Writings of Maimonides, the Moshe Wilensky Archive, the Nuremberg Mahzor,

in 2017) is the NLI Ktiv: The International Collection of Digitized Hebrew Manuscripts (ktiv.en.nli.org.il). The thousands of Hebrew manuscripts that have survived until the present day represent one of the most important sources of Jewish cultural heritage and shed light on the intellectual, religious, and daily life of Jews throughout the ages. Over the centuries, these manuscripts were scattered throughout the world; today, they can be found among numerous libraries, museums, galleries, archives, and private collections. While many original Hebrew manuscripts remain dispersed, copies can be found at the NLI in Jerusalem.

David Ben-Gurion, Israel's first prime minister, had the initiative and vision to gather in Jerusalem as many Hebrew manuscripts as possible. His goal was to create a comprehensive collection that would be used for academic research and religious and cultural enrichment. Once it became clear that it would be impossible to physically collect all these manuscripts, Ben-Gurion decided to establish an institute where photographs of them would be stored, and in 1950, the Institute of Microfilmed Hebrew Manuscripts was established. Today, the institute is an integral part of the National Library and has managed to collect microfilm images of approximately 95 percent of all the Hebrew manuscripts identified in the world. The catalog of Hebrew manuscripts in the NLI, with its seventy thousand to eighty thousand volumes that range from ancient manuscripts dating to before the tenth century to modern manuscripts from the twentieth century, represents the largest and most important surviving "virtual collection" of Hebrew manuscript metadata.²

Considerable information can be extracted from the metadata of the manuscripts, as shown in previous studies.³ However, the manuscripts' textual

the Worms Mahzor, the Rashi, Alexandrovich Collection (<https://www.nli.org.il/en/at-your-service/who-we-are/projects/shapell-digitization>).

2 Malachi Beit-Arié, *Hebrew Codicology: Historical and Comparative Typology of Medieval Hebrew Codices Based on the Documentation of the Extant Dated Manuscripts until 1540 Using a Quantitative Approach* (Jerusalem: Israel Academy of Sciences and Humanities, 2021), <https://doi.org/10.25592/uhhfdm.9349>; Benjamin Richler, "Lezione Dottorale," *Materia Giudaica* 15, nos. 15/16 (2011): 14–16; Colette Sirat, *Hebrew Manuscripts of the Middle Ages* (Cambridge: Cambridge University Press, 2002).

3 Maayan Zhitomirsky-Geffet and Gila Prebor, "Toward an Ontopedia for Historical Hebrew Manuscripts," *Frontiers in Digital Humanities* 3 (March 2016), <https://doi.org/10.3389/fdigh.2016.00003>;

content, while of great importance, remains inaccessible. With significant advances in automatic reading technologies in recent years, it is now possible to explore the potential of using these technologies in combination, working with the scanned images of the manuscript collection in the National Library. This process will create a real revolution in the study of Hebrew manuscripts, the study of the Hebrew text, and Jewish studies in general.

The purpose of this article is to discuss the potential inherent in the use of technological tools in the transcription of Hebrew manuscripts. The paper is structured as follows: First, I give a brief overview of automatic text recognition in general and in Hebrew letters specifically. This is followed by a review of the Transkribus platform and the models that I used and trained using the Transkribus platform with a Sephardic semicursive fifteenth-century script. I then analyze their performance, comparing a generic model to a specific model. I conclude with a discussion of the implications of Transkribus's capabilities for automatic transcription of historical handwritten Hebrew manuscripts.⁴

Automatic Text Recognition (ATR)

The digitization of historical resources is accomplished by professionally photographing and scanning the various resources. This digitization makes historical documents accessible and contributes to the preservation of

Gila Prebor, Maayan Zhitomirsky-Geffet, and Yitzchak Miller, "A Multi-dimensional Ontology-Based Analysis of the Censorship of Hebrew Manuscripts," *Digital Humanities Quarterly* 14, no. 1 (2020): 1–21; Gila Prebor, Maayan Zhitomirsky-Geffet, and Yitzchak Miller, "A New Analytic Framework for Prediction of Migration Patterns and Locations of Historical Manuscripts Based on Their Script Types," *Digital Scholarship in the Humanities* 35, no. 2 (2020): 441–58, <https://doi.org/10.1093/LLC/FQZ038>.

⁴ In October 2022, an announcement was made regarding the allocation of a €10 million grant by the EU European Research Council. The grant was awarded to fund a study entitled "Migrations of Textual and Scribal Traditions via Large-Scale Computational Analysis of Medieval Manuscripts in Hebrew Script," which focuses on Hebrew script and is dedicated to the field of Hebrew text recognition. The anticipated outcomes of this research endeavor hold significant promise for advancing the field of Jewish studies and opening up new avenues of exploration; https://erc.europa.eu/sites/default/files/2022-10/erc_2022_syg.pdf.

knowledge in an electronic form for future generations. But scanning alone is not enough to make these documents accessible in their entirety, since it provides only an image of the scanned page. It does not allow for searching, copying, and analyzing the text found in the scanned document. For these purposes, the scan must be converted from an image format to machine-readable form.

This process of turning an image into a machine-readable form has been already available since the mid-1960s and has made considerable progress in recent decades.⁵ The process usually involves three steps: (1) layout analysis (identifying text regions, finding lines of text, attributing semantic tags to the lines and the regions), (2) simultaneous character recognition and word recognition, and sometimes (3) post-processing with language models adapted to ensure the reliability and quality of the prediction.⁶

The success of the automatic text recognition (ATR) depends on many factors, including the quality of the scan, the quality of the printing, the type of text, the type of font, and the language of the text. The precision of the ATR is measured by the character error rate (CER) and the word error rate (WER) in the text. The fewer errors there are in the translation of the image into text, the higher the quality and accuracy of the resulting text.

A distinction must be made between the processes for handwritten texts and printed texts. The process for the former is referred to as handwritten text recognition (HTR); the process for the latter is called optical text recognition (OCR). Today, OCR systems are capable of recognizing text with a very high degree of accuracy.⁷ HTR software is still under development, and it faces additional challenges compared to OCR: the uniqueness

5 Deepa Berchmans and S. S. Kumar, "Optical Character Recognition: An Overview and an Insight," in *2014 International Conference on Control, Instrumentation, Communication and Computational Technologies, ICCICCT* (Kanyakumari, India: Institute of Electrical and Electronics Engineers, 2014), 1361–65, <https://doi.org/10.1109/ICCICCT.2014.6993174>.

6 Calfa Blog, "How AI can help researchers to transcribe their manuscripts (HTR)?," <https://calfa.fr/blog/28>.

7 Romero, Veronica, Nicolás Serrano, Alejandro H. Toselli, Joan Andreu Sanchez, and Enrique Vidal, "Handwritten Text Recognition for Historical Documents" (paper given at the Language Technologies for Digital Humanities and Cultural Heritage Workshop, Hissar, Bulgaria, 16 September 2011), 90–96.

of handwriting style; variations of handwriting forms; and use of abbreviations, erasures, blacking out, and more.⁸

HTR and OCR in Hebrew Letters

To produce accurate transcriptions of printed text in English, various computer software options have been available for many years. These tools, such as the free open-source software Tesseract and Google Docs, as well as the commercial software ABBYY, have proven to be effective in handling this type of text. However, even ABBYY, which can handle printed modern Hebrew text, lacks the necessary sophistication to accurately transcribe historical materials, particularly handwritten manuscripts.

The Hebrew language presents unique challenges due to several distinct characteristics that set it apart from Latin-based languages. They include the use of different characters, right-to-left writing direction as opposed to left-to-right, and a distinctive morphological and orthographical structure.⁹ Hebrew exhibits a richer morphology compared to English, with linguistic estimates indicating approximately seventy million valid (inflected) forms in Hebrew, whereas English has only one million.¹⁰ In rabbinic literature, there are approximately seventeen thousand different abbreviations, 35 percent of which are ambiguous.¹¹ These challenges are further amplified when dealing with manuscripts.

8 Régis Schlagdenhauffen, “Optical Recognition Assisted Transcription with Transkribus: The Experiment concerning Eugène Wilhelm’s Personal Diary (1885–1951),” *Journal of Data Mining & Digital Humanities*, 28 August 2020, <https://doi.org/10.46298/jdmhdh.6249>.

9 Gila Prebor, “New Technologies for the Collation of Hebrew Texts,” *Zutot* 10, no. 1 (2013): 53–64; Hadar Miller, Moshe Lavee, Nachum Dershowitz, Samuel Londner, and Tsvi Kuflik, “Systematic Textual Availability of Manuscripts,” 2022, <https://api.semanticscholar.org/CorpusID:251919736>.

10 Yaacov Choueka, Ehud S. Conley, and Ido Dagan, “A Comprehensive Bilingual Word Alignment System: Application to Disparate Languages: Hebrew and English,” *Parallel Text Processing: Alignment and Use of Translation Corpora*, ed. Jean Veronis (Dordrecht: Springer Science+Business Media, 2000), 69–96.

11 Yaakov HaCohen-Kerner, Ariel Kass, and Ariel Peretz, “Baseline Methods for Automatic Disambiguation of Abbreviations in Jewish Law Documents” (paper given at *Advances in Natural Language Processing*, 4th International Conference, EsTAL 2004, Alicante, Spain, 20–22 October 2004), 58–69.

One of the prominent approaches in text processing in recent years is natural language analysis and machine learning techniques, such as neural networks. A deep learning approach is based on neural networks that learn different conversion patterns between input and output. The disadvantage of the neural network is the huge amount of manually labeled data required for training, which are often not available. This is also the reason why, until recently, these methods were not used to convert text in Hebrew letters, as there are not sufficient target data to train neural networks in text in Hebrew letters.¹² The exponential increase in digitized materials has led to many efforts to create accessible texts in a machine-readable form; in fact, several large projects have achieved success in this area.

In the past decade, several projects have been developed with the aim of building a web-based infrastructure capable of successfully converting historical manuscripts with different writing systems. This article specifically concentrates on Transkribus.¹³

THE TRANSKRIBUS PLATFORM

Transkribus is a comprehensive platform for the digitization, artificial intelligence-powered text recognition, transcription, and search of historical documents from any place, any time, and in any language (<https://readcoop.eu/transkribus>). Transkribus was launched in 2015 with funding from the European Commission's Seventh Framework Programme (FP7), as part of the tranScriptorium Project (2013–15; <https://eadh.org/projects/transcriptorium>).¹⁴ Transkribus enables the downloading of documents and

12 Omri Suissa, Maayan Zhitomirsky-Geffet, and Avshalom Elmalech, "Toward a Period-Specific Optimized Neural Network for OCR Error Correction of Historical Hebrew Texts," *Journal on Computing and Cultural Heritage* 15, no. 2 (2022): 1–20; <https://doi.org/10.1145/3479159>.

13 An example of additional software is eScriptorium; Benjamin Kiessling, Robin Tissot, Peter Stokes, and Daniel Stökl Ben Ezra, "EScriptorium: An Open Source Platform for Historical Document Analysis," *2019 International Conference on Document Analysis and Recognition Workshops, ICDARW* (2019): 19–24. <https://doi.org/10.1109/ICDARW.2019.10032>.

14 Guenter Muehlberger, Louise Seaward, Melissa Terras, Sofia Ares Oliveira, Vicente Bosch, Maximilian Bryan, Sebastian Colutto, Hervé Déjean, Markus Diem, Stefan Fiel, Basilis Gatos,

images within a private but shareable collection. Using layout analysis tools, Transkribus segments the downloaded images into blocks, then lines, and finally words. Text can be linked to images thus transcribed into any language with the character font chosen. Ultimately, the software allows exportation of transcribed documents in several formats (DOCX, TEI, RTF, PDF, XML).¹⁵

Users work with Transkribus to create ground truth data suitable for machine learning. “Ground truth” is a term commonly used in machine learning to refer to accurate, objective information provided by empirical, direct processes, rather than that inferred from sources via the statistical calculation of uncertainty. In the case of Transkribus, ground truth information is gathered by training the system with enough data (usually around fifteen thousand words, or seventy-five pages) relating to an individual script. These can then be used to create a model that can be applied successfully to large volumes of the same script from submitted images and transcripts. Transkribus engines learn to decipher (historical) handwritten or printed text from digital images and can then automatically generate transcripts of similar material.¹⁶

Transkribus offers a few public models to users, every model with a short description of the training material, which languages the model can be useful for, and who has created and trained it.¹⁷ Transkribus users can benefit from

Albert Greinöcker, Tobias Grüning, Guenter Hackl, Vili Haukkovaara, Gerhard Heyer, Lauri Hirvonen, Tobias Hodel, Matti Jokinen, Philip Kahle, Mario Kallio, Frederic Kaplan, Florian Kleber, Roger Labahn, Eva Maria Lang, Sören Laube, Gundram Leifert, Georgios Louloudis, Rory McNicholl, Jean-Luc Meunier, Johannes Michael, Elena Mühlbauer, Nathanael Philipp, Ioannis Pratikakis, Joan Puigcerver Pérez, Hannelore Putz, George Retsinas, Verónica Romero, Robert Sablatnig, Joan Andreu Sánchez, Philip Schofield, Giorgos Sfikas, Christian Sieber, Nikolaos Stamatopoulos, Tobias Strauß, Tamara Terbul, Alejandro Héctor Toselli, Berthold Ulreich, Mauricio Villegas, Enrique Vidal, Johanna Walcher, Max Weidemann, Herbert Wurster, and Konstantinos Zagoris, “Transforming Scholarship in the Archives through Handwritten Text Recognition: Transkribus as a Case Study,” *Journal of Documentation* 75, no. 5 (2019): 954–76, <https://doi.org/10.1108/JD-07-2018-0114>.

15 Schlagdenhauffen, “Optical Recognition Assisted Transcription with Transkribus.”

16 Muehlberger, “Transforming Scholarship,” 957.

17 “Public AI Models in Transkribus,” READ Coop, <https://readcoop.eu/transkribus/public-models/>.

the network effect and save work and time. The DiJeSt model is a publicly available model that is specifically tailored to handle languages written in Hebrew letters.¹⁸ It is important to note that this model was primarily developed based on printed materials, encompassing a comprehensive “golden” corpus comprising more than 1,800 pages. This corpus consists of diverse printed texts in languages such as Yiddish, Hebrew, Jewish Arabic, and Ladino from the sixteenth through twentieth centuries.¹⁹ Another public model tailored to Hebrew alphabet is the DYBBUK model, prepared by the DYBBUK project team.²⁰ Ground truth data consist mainly of plays in Yiddish handwriting from the late nineteenth and early twentieth centuries written in New York and Europe.²¹

TRANSCRIPTION OF A SEPHARDIC SEMICURSIVE FIFTEENTH-CENTURY SCRIPT

To assess the capabilities of technological tools in transcribing Hebrew manuscripts, a decision was made to employ the Transkribus software on a specific sample. The chosen example was a Sephardic semicursive handwritten text dating back to the fifteenth century. The manuscript selected for this purpose was *Sefer ha-Iqqarim* by Joseph Albo, which is housed in the Bibliothèque nationale de France (BnF) and bears the manuscript identifier MS hébreu 740. It is noteworthy that this manuscript, penned in 1454, originates from Laguna de Negrillos, located in northwest Spain.²² A digital copy

18 Following various updates and modifications implemented in Transkribus, the model formerly known as DiJeSt has been rebranded as the DiJeSt 2.0 model (as of May 2023). “Hebrew Script Languages,” READ Coop, <https://readcoop.eu/model/hebrew-script-languages/>.

19 “Our Text Recognition Ground Truth and Model,” DiJeSt, <http://dijest.net/gtmodel/>.

20 <https://www.dybbuk.co>.

21 Ruthie Abeliovich, “The DYBBUK Model: A New Yiddish Handwriting Text Recognition Tool,” *geveb*, February 2022, accessed 13 July 2023, <https://ingeveb.org/pedagogy/the-dybbuk-model-a-new-yiddish-handwriting-text-recognition-tool>.

22 Malachi Beit-Arié and Edna Engel, *Specimens of Mediaeval Hebrew Scripts*, vol. 2: *Sephardic Script* (Jerusalem: Ha-Akademiyah Ha-le’umit Ha-Yisre’elit Le-mada’im, 2002), 112.

of the manuscript was downloaded from the Ktiv website and subsequently uploaded to the Transkribus Expert Client platform. As mentioned, Transkribus currently lacks a specific model for the ancient Hebrew script. In order to proceed with the transcription, the available Hebrew text recognition public model called DiJeSt was utilized. This particular model had been trained on a combination of historic Hebrew scripts and languages from the twentieth century within the framework of the DiJeSt project (<http://dijest.net/gtmodel>). Despite being aware of the model's limitations, as it had not been specifically trained on Hebrew handwritten texts, it was employed to explore the capabilities of the Transkribus tool in the absence of a more suitable alternative. The transcription obtained from this process is illustrated in figure 1. As anticipated and demonstrated in figure 2, a meticulous examination of the transcription of folio 19r from the *Sefer ha-Iqqarim* manuscript MS hébreu 740, employing the DiJeSt model, revealed numerous

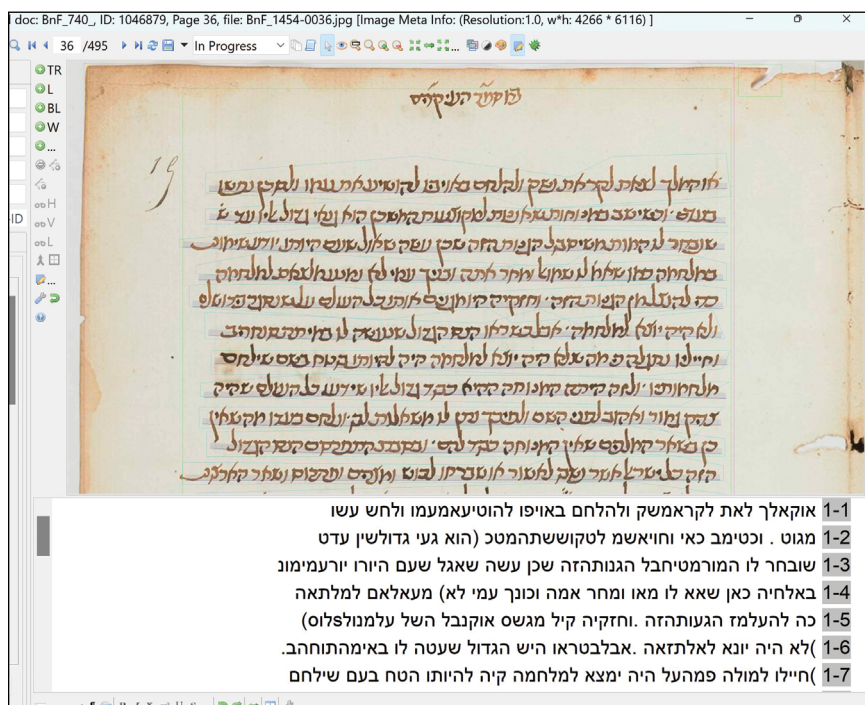
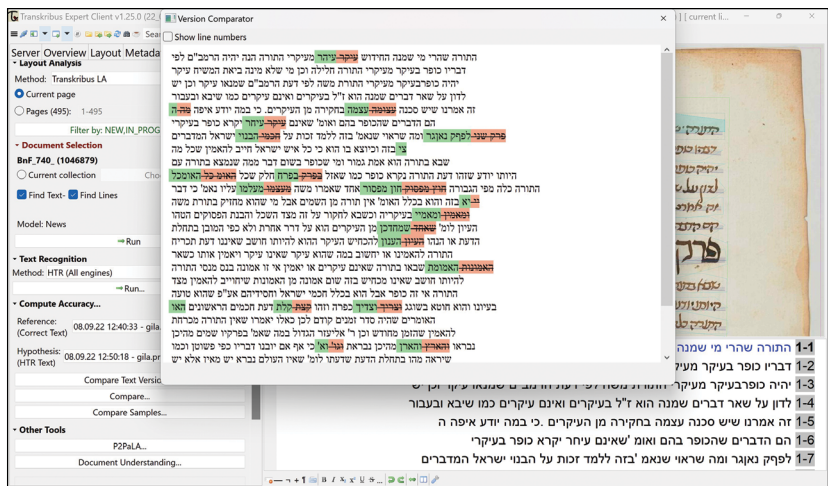


FIGURE 1. Example of transcription of the *Sefer ha-Iqqarim* manuscript, Paris, Bibliothèque nationale de France, MS hébreu 740, fol. 19r, using the DiJeSt model.



MSS	Model	CER	WER
Paris, BnF MS hébreu 740, fol. 19r	DiJeSt model	32.10%	79.34%

FIGURE 2. Comparison of the transcription of *Sefer ha-Iqqarim* manuscript using the DiJeSt model and the correct text.

discrepancies when compared to the accurate text. The evaluation of the transcription's quality unveiled a remarkably high CER of 32.10 percent, accompanied by a WER of 79.34 percent. To improve the model, manual transcription was undertaken for five pages of the folios 13r–15r. A new model was trained using folios 13r–14v for training and folio 15r for validation (1,307 words, 110 lines), with the DiJeSt model used as base model. I named the new mode the BnF 740 model. When I used the new model to read folio 15v, the results improved significantly, as can be seen in figures 3 and 4.

Encouraged by the significant improvement observed, the training of the model was further extended. A broader set of manuscript pages—namely, folios 13r–22v and 62r, spanning a total of twenty pages—were employed for training purposes (fig. 5). This training dataset encompassed 7,641 words across 597 lines. Additionally, folio 23r was designated for validation during this process. Similar to the previous iterations, the DiJeSt model was utilized as the foundational base model. The resulting model was given the name “Heb_Sefaradic_Semi_Cursive_15,” reflecting its focus on the specific script and style of the Sephardic semicursive text from the fifteenth century.

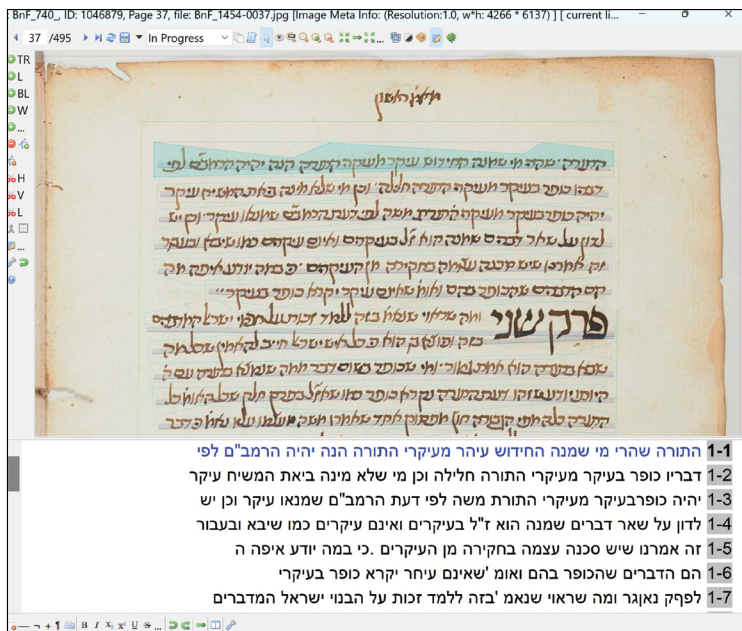
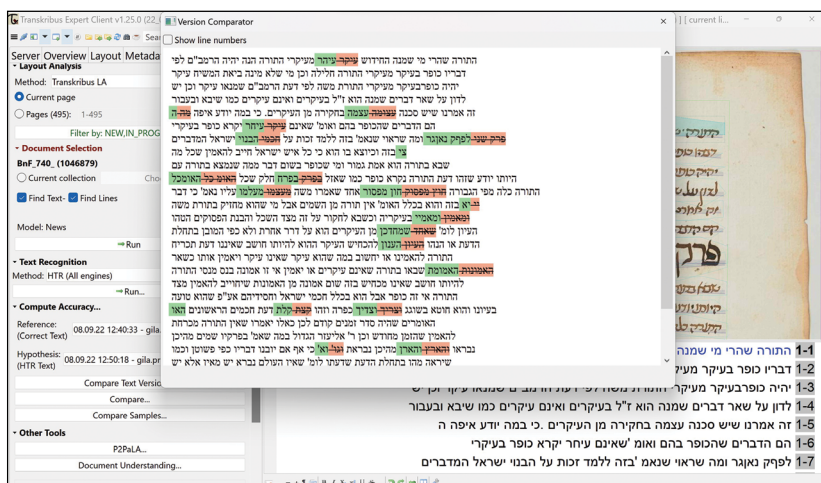


FIGURE 3. Example of a transcription of BnF MS hébreu 740, fol. 15v, using the BnF 740 model.



MSS	Model	CER	WER
Paris, BnF MS hébreu 740, fol. 15v	BnF 740A model	2.45%	7.49%

FIGURE 4. Comparison of the transcription of BnF MS hébreu 740, fol. 15v using the BnF 740A model and the corrected text.

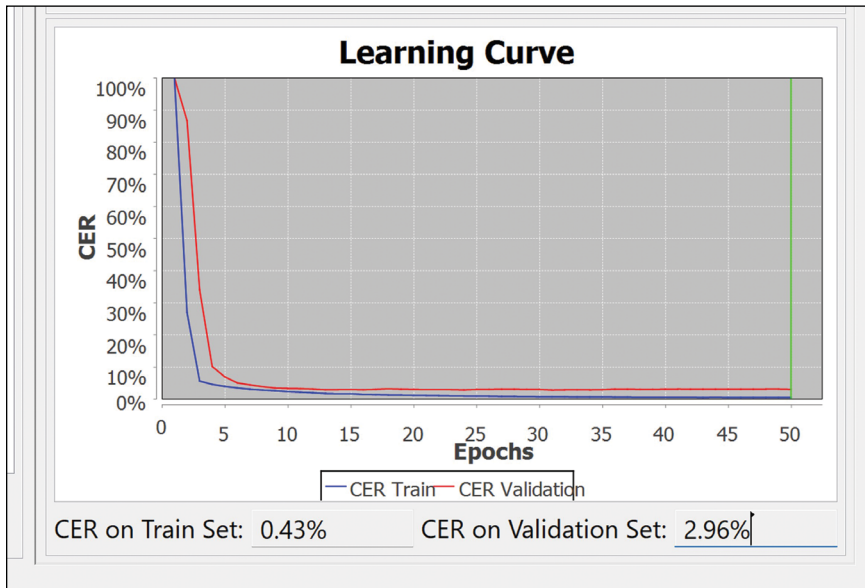
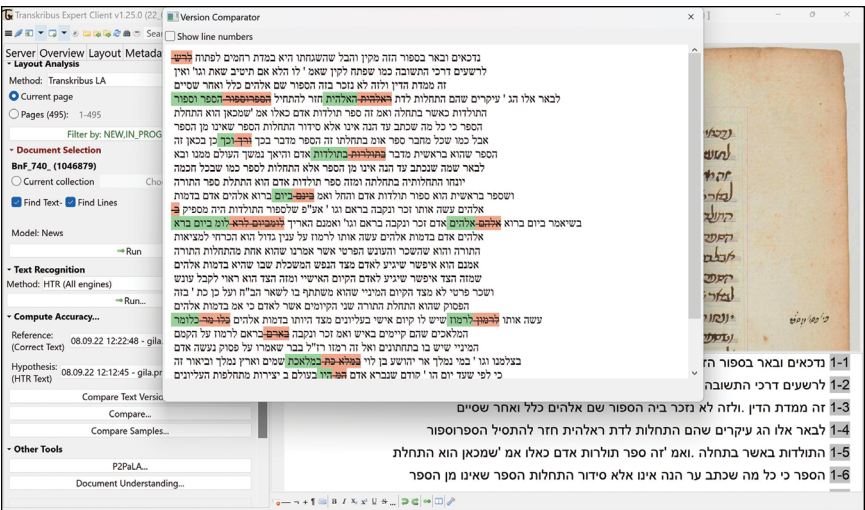


FIGURE 5. The learning curve of Heb_Sefaradic_Semi_Cursive_15 yielded a CER of 0.43 percent on the training set and of 2.96 percent on the test set after fifty epochs (training sessions) of training, meaning that there was fewer than one incorrectly transcribed character for every one hundred characters.

The newly developed model, Heb_Sefaradic_Semi_Cursive_15, was employed to transcribe folio 26v. The outcomes, as depicted in figure 6, were notably impressive. The error rates exhibited a significant reduction, with a CER of merely 1.25 percent and a WER of 5.84 percent (fig. 6).

PITFALLS AND POSSIBLE SOLUTIONS

Figure 7 shows the transcription of MS hébreu 740, folio 26v, using the Heb_Sefaradic_Semi_Cursive_15 model. As can be seen, most of the letters are rendered correctly, even letters connected as in line 4—עיקרים—and line 6—ני. But there are still problems that cannot be ignored, such as extra letters at the end of the line (as seen at the end of line 1). Most of the Hebrew copyists of the Middle Ages strove to produce lines that were as uniform in length



<i>MSS</i>	<i>Model</i>	<i>CER</i>	<i>WER</i>
Paris, BnF MS hébreu 740, fol. 26v	Heb_Sefaradic_Semi_Cursive_15	1.25%	5.84%

FIGURE 6. Comparison of the transcription of BnF MS hébreu 740, fol. 26r, using the Heb_Sefaradic_Semi_Cursive_15 model and the manually corrected text.

as possible. The right margin in Hebrew handwritten books was not troublesome, but the left margin could pose a problem. To overcome that, the Hebrew copyists developed a wide variety of solutions. One of the line-keeping devices was filling the space left at the end of the line by inscribing as many letters of the last word as the boundary margin allowed, then rewriting the entire word at the beginning of the following line, as can be seen in line 1.²³ The letters at the end of line 1—לרש—were used only to keep the left margin straight, and the entire word לרשעים was written in line 2. This is crucial information, and a reader unfamiliar with this method could misunderstand and inaccurately transcribe the text at the end of the lines. Other problems that need to be solved are the identification of various phenomena in the text such as deletions, the integration of footnotes, headings, initial words, running titles at the heads of pages or columns, and catchwords and counter-catchwords in the manuscript.

23 Beit-Arié, *Hebrew Codicology*, ch. 7.

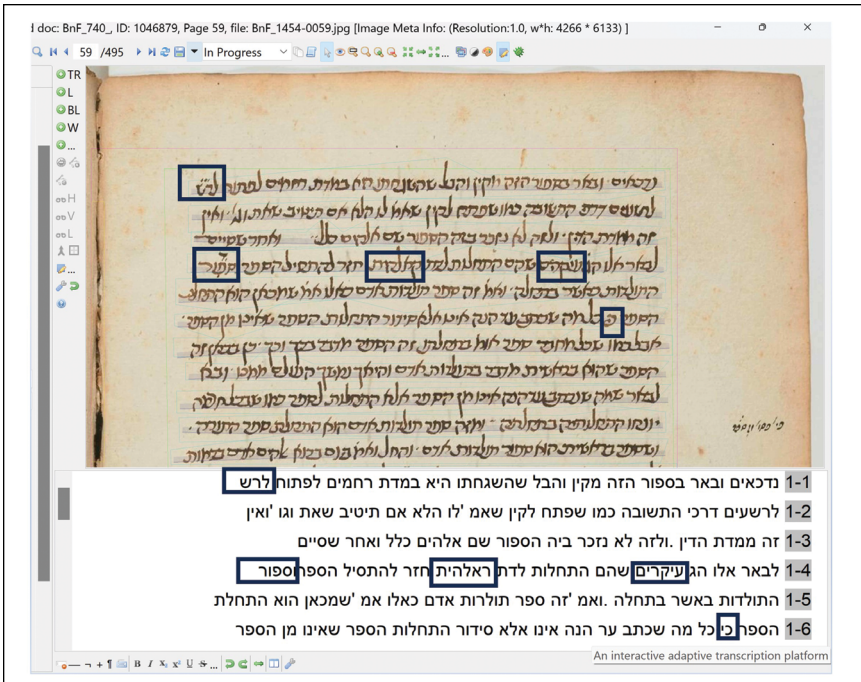


FIGURE 7. Example of transcription of BnF MS hébreu 740, fol. 26v, using Heb_Sefaradic_Semi_Cursive_15 model.

A GENERIC MODEL?

MS hébreu 740, with its 454 pages, has presented various challenges during the transcription process. However, the examples provided above have demonstrated that despite these obstacles, Transkribus has yielded a high level of accuracy and showcased its efficacy in facilitating transcription tasks in this particular case. Moving forward, the next inquiry I sought to address was whether the model trained on a specific manuscript could serve as a generic model for transcribing other manuscripts of the same script type, namely the Sephardic semicursive script from the fifteenth century. To evaluate the applicability of the Heb_Sefaradic_Semi_Cursive_15 model to transcribe additional manuscripts, I specifically selected manuscripts from the fifteenth century that were written in the Sephardic semicursive script

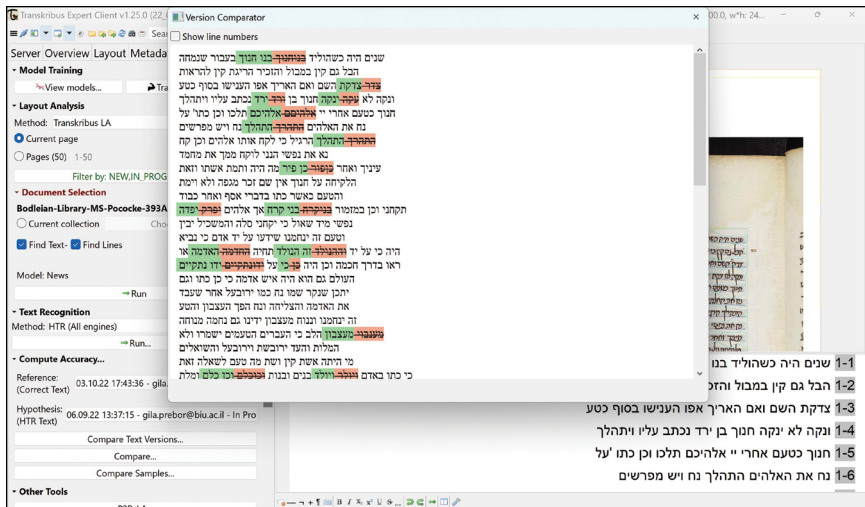


FIGURE 8. Example of transcription of Oxford, Bodleian Library, MS Poc. 393, fol. 11r, using Heb_Sefaradic_Semi_Cursive_15.

and had available digital files. These manuscripts were chosen from the collection featured in *Specimens of Mediaeval Hebrew Scripts*, volume 2 *Sephardic Script*.²⁴

Manuscript Oxford, Bodleian Library, MS Poc. 393 (F 16353), contains commentary on the Pentateuch by Abraham ibn Ezra, which was written in Sephardic semicursive script probably in Provence in 1445.²⁵ I uploaded it to the Transkribus platform and employed the Heb_Sefaradic_Semi_Cursive_15 model for the transcription process. The outcome of this transcription can be observed in figure 8, which displays the resulting text.

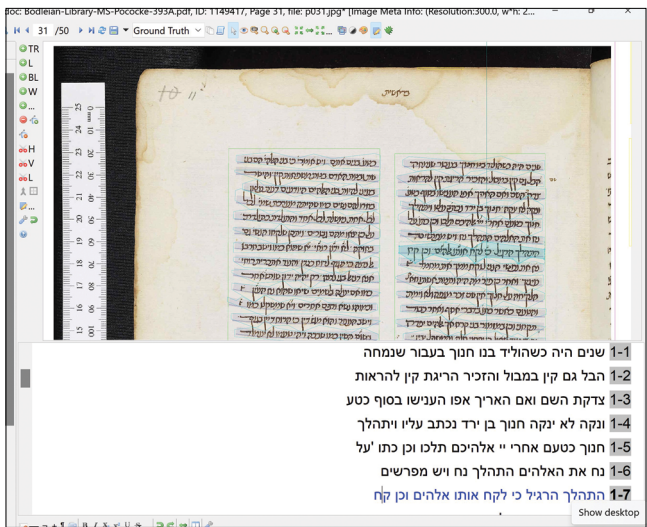
There are several errors in the transcription, but there are also several difficult cases where the software can be seen to have handled the difficulties well. For example, in line 7, the beginning of the word אלהים is written as a ligature and the letters א and ל are joined together into a single glyph; in the transcription, the letters were correctly separated. In line 6, there is a graphic mark to complete the line, and the software ignored it in the transcription.

24 Beit-Arié and Engel, *Specimens of Mediaeval Hebrew Scripts*, vol 2.

25 Beit-Arié and Engel, *Specimens of Mediaeval Hebrew Scripts*, 2:108.

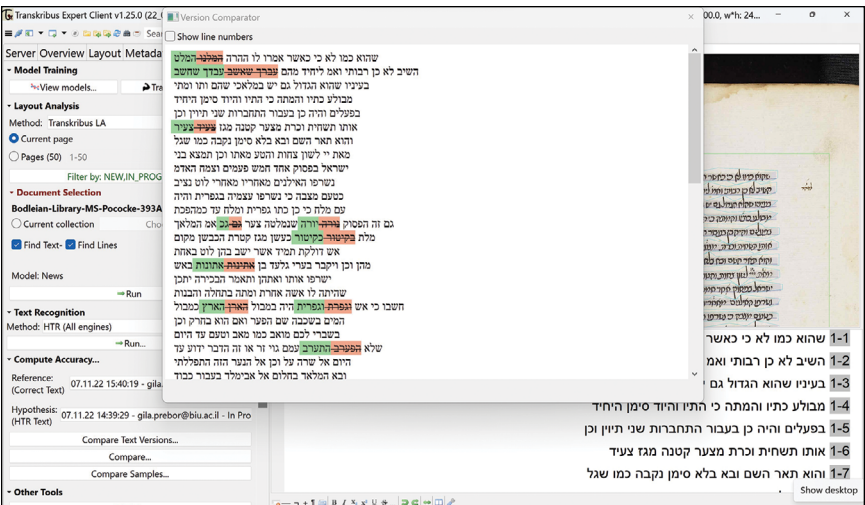
In line 9, the letter ן in the last word is written above the line and here, the transcription is correct. As can be seen in figure 8, the percentage of errors in folio 11r is reasonable (mainly in relation to the effort invested)—2.29 percent CER and 11.21 percent WER. When I tested the model on additional pages, the results were similar, and the percentage of errors on average was 3.31 percent CER and 14.79 percent WER (fig. 9).

To investigate potential improvements in transcription results, I conducted an additional training session using the Heb_Sefaradic_Semi_Cursive_15 model. This training session involved a dataset consisting of six pages, with an additional page reserved for validation purposes. The training set contained a total of 2,786 words and 347 lines, with the Heb_Sefaradic_Semi_Cursive_15 model used as base model. The improved model is called Oxford_393 model.



MSS	Model	CER	WER
Oxford, BL Poc. 393, fol. 11r	Heb_Sefaradic_Semi_Cursive_15	2.29%	11.21%
Oxford, BL Poc. 393, fol. 12r	Heb_Sefaradic_Semi_Cursive_15	2.68%	11.21%
Oxford, BL Poc. 393, fol. 12v	Heb_Sefaradic_Semi_Cursive_15	3.78%	16.09%
Oxford, BL Poc. 393, fol. 13r	Heb_Sefaradic_Semi_Cursive_15	3.83%	16.92%
Oxford, BL Poc. 393, fol. 13v	Heb_Sefaradic_Semi_Cursive_15	3.98%	18.52%
Average	3.31%	14.79%	

FIGURE 9. Comparison of the transcription of Oxford, Bodleian Library, MS Poc. 393, fol. 11r, using Heb_Sefaradic_Semi_Cursive_15 model and the corrected text.



<i>MSS</i>	<i>Model</i>	<i>CER</i>	<i>WER</i>
Oxford, BL. Poc. 393, fol. 20r	Oxford_393	1.27%	5.97%

FIGURE 10. Comparison of the transcription of the Oxford, Bodleian Library, MS Poc. 393, fol. 19v, using the Oxford_393 model and the corrected text.

Figure 10 illustrates the notable improvement achieved through this model. The CER reduced to an impressive 1.27 percent, and the WER decreased to 5.97 percent. These figures indicate a substantial enhancement in accuracy compared to the previous transcription results.

The success of the Oxford_393 model highlights the potential for improving transcription outcomes by carefully selecting and training models with specific datasets. This focused approach can effectively address the challenges posed by Sephardic semicursive script and lead to significantly improved transcription results.

Conclusion

This article has demonstrated the immense potential of technological tools in the transcription of Hebrew manuscripts. Through my analysis, I have shown that handwriting recognition models trained with Transkribus can generate usable results when applied to Hebrew Sephardic semicursive manuscripts from the fifteenth century. This marks a significant advancement

in the field, as it allows for a more efficient and cost-effective approach to transcription.

By leveraging the power of machine learning, computers can now produce a first draft of a transcription, which, although requiring manual correction, significantly reduces the time and resources needed compared to manual transcription from scratch. This breakthrough has far-reaching implications, as it opens up avenues for handling large volumes of manuscripts in a shorter period, enabling researchers and institutions to tackle previously daunting tasks.

Furthermore, my findings highlight that even with a relatively small investment, remarkable results can be achieved. The recommended amount of ground truth data for training a Transkribus model is set at approximately fifteen thousand transcribed words, or seventy-five pages, to recognize text written by a single hand. Adhering to the principles of machine learning, the submission of a larger volume of ground truth data enhances the accuracy of the transcription results. However, our trials have shown that even with a smaller amount of data, it is still possible to attain good outcomes. This is a promising prospect, as it facilitates the mass digitization of previously unpublished manuscripts, opening up vast opportunities for future research endeavors.

The horizon holds immense potential for the continued development of such tools and the exploration of additional techniques to enhance the transcription accuracy and efficiency of Hebrew manuscripts.